



PRECURSOR ANALYSIS REPORT: SQL SLAMMER WORM INFECTION OF DAVIS-BESSE NUCLEAR POWER PLANT 2003

Cybersecurity for the Operational Technology
Environment (CyOTE)

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PRECURSOR ANALYSIS REPORT: SQL SLAMMER WORM INFECTION OF DAVIS-BESSE NUCLEAR POWER PLANT 2003

1. EXECUTIVE SUMMARY

The SQL Slammer Worm Infection of Davis-Besse Nuclear Power Plant 2003 Precursor Analysis Report leverages publicly available information about Davis-Besse's 2003 cyber attack and catalogs anomalous observables for each technique employed in the attack. This analysis is based upon the methodology of the Cybersecurity for the Operational Technology Environment (CyOTE) program.

On 25 January 2003, the SQL Slammer worm infected more than 90% of vulnerable hosts and crashed the internet in 10 to 15 minutes, making it one of the fastest spreading worms in history. SQL Slammer is a fileless, memory-resident worm that remotely exploits a stack-based buffer overflow vulnerability on local hosts to intensively scan and rapidly self-propagate across the internet. The worm infected approximately 300,000 unpatched hosts running Microsoft Structured Query Language (SQL) Server 2000 or Microsoft Desktop Engine (MSDE) 2000 with SQL Server Resolution Service.

The SQL Slammer worm indirectly infected FirstEnergy's Davis-Besse nuclear power plant by first infecting a consultant's company network server and then propagating through an external misconfigured connection into Davis-Besse's site network. The infection caused major network congestion, slow performance, data overloads, and the inability of local hosts to communicate with each other, which eventually caused a loss of availability and a loss of view when the Safety Parameter Display System (SPDS) and Plant Process Computer (PPC) crashed. At the time of the infection, the plant was already offline, the digital monitoring systems had redundant analog backups, and the plant control and safety functions were not affected, so there were no concerns of a safety breach. However, this incident resulted in many lessons learned and spawned important discussions about cybersecurity's role in nuclear safety and electric power reliability regulation, policy, and guidance.

Researchers and analysts identified 10 unique techniques utilized during the attack with a total of 640 observables using MITRE ATT&CK® for Industrial Control Systems. The CyOTE program assesses observables accompanying techniques used prior to the triggering event to identify opportunities to detect malicious activity. If observables accompanying the attack techniques are perceived and investigated prior to the triggering event, earlier comprehension of malicious activity can take place. Eight of the identified techniques used during Davis-Besse cyber attack were precursors to the triggering event. Analysis identified 596 observables associated with these precursor techniques, 428 of which were assessed to have an increased likelihood of being perceived in the 331 days preceding the triggering event. The response and comprehension time could have been reduced if the observables had been identified earlier.

The information gathered in this report contributes to a library of observables tied to a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack. Asset owners and operators can use these products if they experience similar observables or to prepare for comparable scenarios.

2. INTRODUCTION

The [Cybersecurity for the Operational Technology Environment \(CyOTE\)](#) program developed capabilities for energy sector organizations to independently identify adversarial tactics and techniques within their operational technology (OT) environments. Led by Idaho National Laboratory (INL) under leadership of the Department of Energy (DOE) Office of Cybersecurity, Energy Security, and Emergency Response (CESER), CyOTE is a partnership with energy sector owners and operators whose goal is to tie the impacts of a cyber attack to anomalies in the OT environment to determine whether the anomalies have a malicious cyber cause.

2.1. APPLYING THE CYOTE METHODOLOGY

The CyOTE methodology, as shown in [Figure 1](#), CyOTE Methodology, applies fundamental concepts of perception and comprehension to a universe of knowns and unknowns increasingly disaggregated into observables, anomalies, and triggering events. The program utilizes MITRE's ATT&CK® Framework for Industrial Control Systems (ICS) as a common lexicon to assess triggering events. By leveraging the CyOTE methodology with existing commercial monitoring capabilities and manual data collection, energy sector partners can understand relationships between multiple observables, which could represent a faint signal of an attack requiring investigation. CyOTE can assist organizations in prioritizing their OT environment visibility investments.

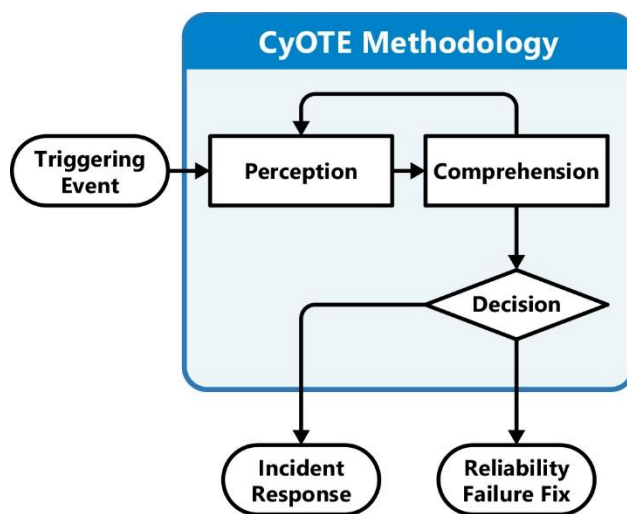


Figure 1. CyOTE Methodology

Historical case studies such as this one support continued learning through analysis of incidents that have impacted OT. This precursor analysis report is based on publicly available reports and provides examples of how key concepts in the CyOTE methodology appear in the real world, providing insights on how similar novel attacks could be detected earlier and therefore mitigated. The analysis enables OT personnel to independently identify observables associated with techniques known to be indicators of attack within OT environments. The identified observables highlight anomalous events for further investigation, which could enhance comprehension of malicious activity.

A timeline of events based on the CyOTE methodology portrays the attack-related observables associated with the precursor analysis report cyber attack. The timeline includes assessed dates, the triggering event, and comprehension of malicious activity by the organization. The point on this timeline when each technique appears is critical to the organization's ability to perceive and comprehend the associated malicious activity. Perception of techniques early in the timeline is critical, since halting those techniques will generally have greater potential to limit additional attack vectors using other techniques, defeat the cyber attack, and limit damage to operations.

Each technique has an assessed perceivability. Perceivability is a function of the number of observables and the potential for personnel to detect those observables. If a technique includes effects which personnel may detect, such as deletion or modification of system files or required user execution, then the technique would be more perceivable.

Differences in infrastructure and system configurations may present different challenges and opportunities for observable detection. For example, architecture-wide endpoint monitoring is likely to improve the perceivability of techniques which modify host files, such as the Data Destruction technique (T0809) for Inhibit Response Function and Theft of Operational Information technique (T0882) for Impact. Network monitoring and log analysis capabilities are likely to improve perceivability of techniques which create malicious network traffic, such as the Standard Application Layer Protocol technique (T0869) for Command and Control, External Remote Services technique (T0822) for Initial Access, and Connection Proxy technique (T0884) for Command and Control. Alternatively, enhancing the monitoring parameters of system files would increase the perceivability of techniques such as Data from Information Repositories technique (T0811) for Collection and the Service Stop technique (T0881) for Inhibit Response Function.

Comprehension can be further enhanced by technique artifacts created when adversaries employ certain attack techniques. The CyOTE program provides organizations with a [library of observables](#) reported in each historical case. The library can be used in conjunction with a repository of artifacts, data sources, and technique detection references for practitioners and developers to support the comprehension of indicators of attack.

2.2. BACKGROUND ON THE ATTACK

The SQL Slammer worm infected more than 90% of vulnerable hosts and crashed the internet in 10 to 15 minutes in late January 2003, making it one of the fastest spreading worms in history.^{1,2} SQL Slammer is a fileless, memory-resident worm that remotely exploits a stack-based buffer overflow vulnerability ([CVE-CAN-2002-0649](#)) on local hosts to intensively scan and rapidly self-propagate.^{3,4,5} The worm infected approximately 300,000 unpatched hosts running Microsoft Structured Query Language (SQL) Server 2000 or Microsoft Desktop Engine (MSDE) 2000 with SQL Server Resolution Service over User Datagram Protocol (UDP) Port 1434.⁶ At certain points in the worm's spread, more than 55 million hosts were scanned every second, with replicated hosts doubling every 8.5 seconds.^{7,8} The global recovery cost is estimated at \$1 billion, and the global cost of lost productivity is estimated at \$1.2 billion.⁹

Among SQL Slammer's victims was FirstEnergy's Davis-Besse nuclear power plant in Ottawa County, Ohio. At 9:00 AM (M-470) on 25 January 2003 (D-0), users on Davis-Besse's enterprise network noticed slow performance, which was about the same time SQL Slammer started infecting internet-connected networks across the world.^{10,11}

At the time of the SQL Slammer infection, Davis-Besse was offline and in a safely defueled condition because a large cavity had been found in the Reactor Pressure Vessel (RPV) head, which averted any major safety issues.¹²

A timeline of adversarial techniques is shown in [Figure 2](#). The timeline includes the estimated number of days prior to and after the triggering event. The timeline after the triggering event includes the assessed victim comprehension timeline.^a

A critical factor in Davis-Besse's infection was the failure of plant personnel to perform a critical software update that was released months earlier. On 10 July 2002 (D-199), Microsoft released a critical security patch/update for Microsoft SQL Server 2000.¹³ On 24 July (D-185), Microsoft published [Security Bulletin MS02-039](#) informing system administrators running Microsoft SQL Server 2000 of a critical buffer overflow vulnerability ([CVE-CAN-2002-0649](#)), found to be exploitable in SQL Server Resolution Service over UDP Port 1434, which the available critical security patch/update would fix.¹⁴ Davis-Besse's IT personnel and plant computer engineers were not aware of the critical security patch/update or bulletin, which directly applied to vulnerable SQL Server 2000 hosts the plant employed in both the enterprise network and operations networks.^{15,16}

The SQL Slammer worm indirectly infected Davis-Besse's site network by first infecting a consultant company's network server, which was using UDP Port 1434 for data transfers, and then propagating

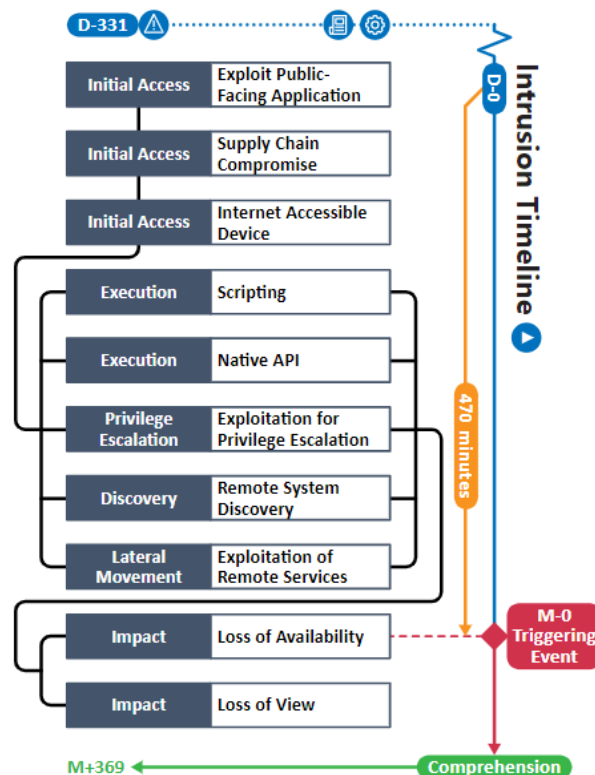


Figure 2. Intrusion Timeline

^a "M" events correspond to minutes prior to (M-) or after (M+) the triggering event (M-0); and "D" events correspond to days prior to (D-) or after (D+) the triggering event (D-0).

through an external misconfigured Transmission System 1 (T1) connection into Davis-Besse's site network.^{b,17} The consultant company accessed the plant's site network via the T1 connection, which itself was implemented behind Davis-Besse's external-facing enterprise firewall, bypassing the firewall's access control policy and ruleset configured to block UDP Port 1434. At least some Davis-Besse IT personnel were aware of this misconfigured connection.¹⁸

In February 2002 (D-331), the Nuclear Regulatory Commission (NRC) issued a security order alerting licensees about external connections that bypass network boundary protections.¹⁹ Davis-Besse's IT personnel addressed the fulfillment of the NRC's security order; however, the external T1 connection used by the consultant, bypassing the external-facing enterprise firewall, remained in place and misconfigured. The relevant plant computer engineers were not informed of the external T1 connection's placement nor the IT personnel's decision to address the NRC's security order.²⁰

Davis-Besse's enterprise network was interconnected with the operations network, with no firewall segregating traffic passing into the operations network, so the worm had unobstructed access to saturate bandwidth and infect vulnerable Microsoft SQL Server 2000 hosts that had UDP Port 1434 open within the enterprise network and operations networks.^{21,22} The resulting intense network scanning and flood of packet clones from SQL Slammer into the network can cause performance issues, increased latency and packet loss, network outages, denial of service, and system crashes.^{23,24,25}

After infecting Davis-Besse's enterprise network, the worm spread to an unpatched SQL Server 2000 host in the operations network, with plant personnel noticing slow performance by 4:00 PM (M-50).^{26,27} At 4:50 PM (M-0), the network congestion caused by SQL Slammer's intensive scanning crashed the Safety Parameter Display System (SPDS), which is the plant's computerized display panel that relays critical plant safety parameters to plant operators and the NRC Operations Center in real time.^c Even though the plant was offline at the time of the infection, many of the safety measures still require diligent monitoring. At 5:13 PM (M+23), the Plant Processing Computer (PPC), which is a plant-wide input/output (I/O) information system for consolidated analog and digital data acquisition and transfer, crashed.^{28,29,30} However, the SPDS and the PPC had redundant analog backups that were unaffected by the worm, and plant control and safety functions were not affected.

Ultimately, the infection of Davis-Besse's site network caused major network congestion, slow performance, data overloads, the inability of local hosts to communicate with each other, and system crashes.^{31,32} It took four hours and 50 minutes to recover the SPDS (M+290) and six hours and nine minutes to recover the PPC (M+369). As the SQL Slammer worm is memory-resident, infected servers were shut down to remove the worm, isolated from the network to prevent reinfection, updated with the available critical security patch/update, and reconnected to the site network. Though there were no concerns of a safety breach, this incident resulted in many lessons learned and spawned important dialogs about cybersecurity's role in nuclear safety and electric power reliability regulation, policy, and guidance.^{33,34}

Analysis identified 10 unique techniques in a sequence and timeframe likely used by adversaries during this cyber attack ([Error! Reference source not found. 1](#)). These attack techniques are defined according to MITRE's ATT&CK® for ICS framework.

^b The "site network" refers to the entirety of the Davis-Besse network.

^c At the time, any SPDS outage lasting eight hours or more required that the NRC be notified.

Table 1. Techniques Used in the SQL Slammer Worm Infection of Davis-Besse Nuclear Power Plant 2003

Initial Access	Execution	Persistence	Privilege Escalation	Evasion	Discovery	Lateral Movement	Collection	Command and Control	Inhibit Response Function	Impair Process Control	Impact
Data Historian Compromise	Change Operating Mode	Modify Program	Exploitation for Privilege Escalation	Change Operating Mode	Network Connection Enumeration	Default Credentials	Automated Collection	Commonly Used Port	Activate Firmware Update Mode	Brute Force I/O	Damage to Property
Drive-by Compromise	Command-Line Interface	Module Firmware	Hooking	Exploitation for Evasion	Network Sniffing	Exploitation of Remote Services	Data from Information Repositories	Connection Proxy	Alarm Suppression	Modify Parameter	Denial of Control
Engineering Workstation Compromise	Execution through API	Project File Infection		Indicator Removal on Host	Remote System Discovery	Lateral Tool Transfer	Detect Operating Mode	Standard Application Layer Protocol	Block Command Message	Module Firmware	Denial of View
Exploit Public-Facing Application	Graphical User Interface	System Firmware		Masquerading	Remote System Information Discovery	Program Download	I/O Image		Block Reporting Message	Spoof Reporting Message	Loss of Availability
Exploitation of Remote Services	Hooking	Valid Accounts		Rootkit	Wireless Sniffing	Remote Services	Man in the Middle		Block Serial COM	Unauthorized Command Message	Loss of Control
External Remote Services	Modify Controller Tasking			Spoof Reporting Message		Valid Accounts	Monitor Process State		Data Destruction		Loss of Productivity and Revenue
Internet Accessible Device	Native API						Point & Tag Identification		Denial of Service		Loss of Protection
Remote Services	Scripting						Program Upload		Device Restart/Shutdown		Loss of Safety
Replication Through Removable Media	User Execution						Screen Capture		Manipulate I/O Image		Loss of View
Rogue Master							Wireless Sniffing		Modify Alarm Settings		Manipulation of Control
Spearphishing Attachment									Rootkit		Manipulation of View
Supply Chain Compromise									Service Stop		Theft of Operational Information
Transient Cyber Asset									System Firmware		
Wireless Compromise											

Table 2. Precursor Analysis Report Quantitative Summary

Precursor Analysis Report Quantitative Summary	Totals
MITRE ATT&CK® for ICS Techniques	10
Technique Observables	640
Precursor Techniques	8
Precursor Technique Observables	596
Highly Perceivable Precursor Technique Observable	428

3. OBSERVABLE AND TECHNIQUE ANALYSIS

The following analysis may assist organizations in identifying malicious cyber activity earlier and more effectively. The following techniques and observables were compiled from publicly available sources and correlated with expert analysis.

3.1. EXPLOIT PUBLIC-FACING APPLICATION TECHNIQUE (T0819) FOR INITIAL ACCESS

At about 9:00 AM on 25 January 2003, the SQL Slammer worm remotely exploited [CVE-CAN-2002-0649](#) on a consultant's public-facing company network server and on unpatched SQL Server 2000 hosts within Davis-Besse's site network.^{35,36,37} This vulnerability allowed a deliberately designed packet, remotely sent to SQL Server Resolution Service over UDP Port 1434, to overwrite portions of system memory, granting remote execution of arbitrary, memory-injected code at an elevated privilege.^{38,39,40} IT personnel and plant computer engineers were not aware of the critical security patch/update, available six months before the infection.^{41,42}

The central exploit of SQL Slammer is the first byte of the UDP packet being a 0x04 flag.^{43,44} Any bytes after the 0x04 flag are input to the Sprintf() function during the SQL Server Monitor thread process of generating a registry key to open. The Sprintf() function takes the input and formats it as output into a fixed-sized destination buffer on the stack frame. The Sprintf() function does not verify nor limit the input sent to write to or overwrite the fixed-sized buffer, and there is no presence of a 0x00 flag byte as an end indicator within the packet, enabling SQL Slammer's code to write outside of the fixed-sized buffer's bounds and perform a stack-based buffer overflow.^{45,46}

IT Staff, IT Cybersecurity, and Support Staff personnel may have been able to observe a critical security patch/update, a Microsoft Security Bulletin, and a Carnegie Mellon Vulnerability Note.

A total of 102 observables were identified with the use of the [Exploit Public-Facing Application technique \(T0819\)](#). This technique is important for investigation because it provides initial access into a public-facing, vulnerable network. This technique appears early in the technique timeline and responding to it will halt all future remote exploitative code execution on hosts with the [CVE-CAN-2002-0649](#) vulnerability. Responding to the critical security patch/update or the Microsoft Security Bulletin would have prevented the SQL Slammer worm from propagating to Davis-Besse's site network. Terminating the chain of techniques at this point would have effectively eliminated the SQL Slammer worm's root means of exploitation for infection.

Of the 102 observables associated with this technique, 74 are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 55 artifacts could be generated by the Exploit Public-Facing Application technique
Technique Observers^d	IT Staff, IT Cybersecurity, Support Staff
Resources	Technique Detection References

^d Observer titles are adapted from the Job Role Groupings listed in the [SANS ICS Job Role to Competency Level Poster](#). CyOTE products utilize these job categories rather than organizational titles to both support comprehensive analysis and preserve anonymity within the victim organization. A complete list of potential observers can be found in [Appendix C](#).

3.2. SUPPLY CHAIN COMPROMISE TECHNIQUE (T0862) FOR INITIAL ACCESS

The SQL Slammer worm propagated to Davis-Besse's site network by first infecting an unnamed consultant's unsecure company network via exploiting a public-facing, unpatched company network server using UDP Port 1434 for data transfers.^{47,48} The worm then propagated through an external misconfigured T1 connection used by the consultant, which bridged the consultant's unsecure company network with Davis-Besse's enterprise network, to further exploit and infect hosts within Davis-Besse's enterprise and operations networks. The T1 connection was used by the consultant to provide an application software that ran on a server in Davis-Besse's site network.

IT Staff, IT Cybersecurity, and Support Staff personnel may have been able to observe the presence of an anomalous T1 connection and unmonitored network traffic. Potential observers may have also been able to observe anomalous system behavior on the host with the T1 connection and on the local area network gateway linking the host with the T1 connection to other networks.

A total of four observables were identified with the use of the [Supply Chain Compromise technique \(T0862\)](#). This technique is important for investigation because it provides initial access into a vulnerable network through an indirect route. This technique appears early in the timeline and responding to it would have limited all future SQL Slammer scanning and propagation from the consultant's company network to Davis-Besse's site network. Terminating the chain of techniques at this point would have effectively eliminated SQL Slammer's point of origination into Davis-Besse's site network.

All four observables associated with this technique are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 31 artifacts could be generated by the Supply Chain Compromise technique
Technique Observers	IT Staff, IT Cybersecurity, Support Staff
Resources	Technique Detection References

3.3. INTERNET ACCESSIBLE DEVICE TECHNIQUE (T0883) FOR INITIAL ACCESS

The misconfigured T1 connection used by the consultant, bridging the consultant's unsecure company network with Davis-Besse's enterprise network, provided a backdoor from the internet to the plant's enterprise network in an unmonitored and unsegregated manner.^{49,50} The T1 connection was implemented behind Davis-Besse's external-facing enterprise firewall, bypassing the external-facing enterprise firewall's access control policy and ruleset configured to block UDP Port 1434. The misconfigured T1 connection caused vulnerable hosts within Davis-Besse's site network to be directly internet reachable. At least some employees of the plant's IT department were aware of the T1 connection, but when the T1 connection was implemented is unknown. The plant's relevant computer engineers were not informed of the external T1 connection's placement nor the decision to address the NRC security order, available 11 months before the infection.⁵¹ Davis-Besse's enterprise network was interconnected with the operations network, and no firewall was segregating traffic passing into the operations network.⁵² Once the worm infected the enterprise network, it had unobstructed access to saturate bandwidth and pseudo-randomly infect vulnerable Microsoft SQL Server 2000 hosts in the operations network that had Port 1434 open.⁵³

IT Staff, IT Cybersecurity, Engineering, and Support Staff personnel may have been able to observe the presence of an anomalous NRC security order, T1 connection, and unmonitored network traffic. Potential observers may have also been able to observe anomalous system behavior on the host with the T1 connection and on the local area network gateway linking the host with the T1 connection to other networks.

A total of five observables were identified with the use of the [Internet Accessible Device technique \(T0883\)](#). This technique is important for investigation because it provides initial access into a vulnerable network through a path that bypasses network boundary protection devices. This technique appears early in the timeline and responding to it would have limited additional SQL Slammer scanning and propagation from the consultant's company network to Davis-Besse's site network. Properly responding to the NRC's security order would have entirely prevented the SQL Slammer worm from infecting Davis-Besse's site network. Terminating the chain of techniques at this point would have effectively eliminated SQL Slammer's open path into Davis-Besse's site network.

All five observables associated with this technique are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 23 artifacts could be generated by the Internet Accessible Device technique
Technique Observers	IT Staff, IT Cybersecurity, Engineering, Support Staff
Resources	Technique Detection References

3.4. SCRIPTING TECHNIQUE (T0853) FOR EXECUTION

SQL Slammer is a Windows-based worm pre-written in x86 assembly code, and it accomplishes the steps of “Get Inside”, “Reprogram the Machine”, “Choose Victims at Random”, “Replicate”, and “Repeat” on every vulnerable host that is reachable and infected by the pseudo-random flood of single 376-byte UDP packets (404-byte packets with the requisite headers) that successive SQL Slammer infections generate.^{54,55,56} The structure of the simple 376-byte packet containing the SQL Slammer worm code is comprised of the enabling vulnerability ([CVE-CAN-2002-0649](#)), the propagation mechanism, and the payload.⁵⁷

SQL Slammer’s code did not carry a directly malicious payload, but rather had the unintended effect of performing denial of service (DoS) attacks due to its rapidly propagated packets saturating available bandwidth.^{58,59} SQL Slammer’s packet content is disguised as a routine query request for SQL Server Resolution Service; however, the packet actually contains a string that is too long for SQL Server Monitor to properly handle.^{60,61} This too-long UDP request string will act on a stack-based buffer overflow to overwrite portions of the system memory with deliberate data as a means to execute arbitrary code in the security context of the domain or SYSTEM level privileges of the SQL Server service account.^{62,63,64} This arbitrary code execution remains in the infected host’s memory and propagates a high volume of SQL Slammer clones by pointing at its own code as the data to send, in 376-byte one-way UDP packets, to hosts with an IP address generated by the worm’s pseudo-random scan cycle instance. If the packet is sent to a vulnerable machine, the machine is infected, and the worm begins to propagate from the newly infected host.⁶⁵

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe anomalous network traffic over UDP Port 1434, network traffic over TCP Port 179, UDP packet contents, and system behavior on local hosts.

A total of 97 observables were identified with the use of the [Scripting technique \(T0853\)](#). This technique is important for investigation because it provides the automated, self-propagating means of deploying and executing a worm. This technique appears midway in the technique timeline and responding to it will limit all future scripted exploitation and propagation techniques on hosts with the [CVE-CAN-2002-0649](#) vulnerability. Terminating the chain of techniques at this point would limit SQL Slammer’s outward propagation, rapid replication rate, and bandwidth saturation.

Of the 97 observables associated with this technique, 69 are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 12 artifacts could be generated by the Scripting technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.5. NATIVE API TECHNIQUE (T0834) FOR EXECUTION

SQL Slammer is a Windows-based worm pre-written in x86 assembly code; as such, SQL Slammer interacts with the Windows operating system (OS) native application programming interfaces (API) to further access and utilize SYSTEM level functions throughout the entirety of the exploitation and propagation steps.^{66,67} Native APIs provide a means of calling low-level hardware, memory space, and process services within the Windows Kernel. This functionality is accessible by user-mode applications and libraries. Native APIs are normally for boot and for carrying out standard tasks and requests, but SQL Slammer uses native APIs to carry out malicious tasks and requests associated with its exploitation and propagation steps.

SQL Slammer's ability to perform the exploit lies within the first byte of the packet, which is a 0x04 byte, and the `Sprintf()` function passing too much data to the fixed-size destination buffer, which allows arbitrary code to be written to and executed in memory.⁶⁸ SQL Slammer's ability to import functions from the `SQLsort.dll`, `ws2_32.dll`, and `kernel32.dll` libraries lies within the `LoadLibrary()` function, along with `get handles` referring to the libraries that functions are imported from.⁶⁹ SQL Slammer's ability to create pseudo-random IP addresses with a starting point value lies within the `GetProcAddress()` and `GetTickCount()` functions. The `GetTickCount()` function will seed each linear congruent pseudo-random number generator (PRNG) scan cycle instance with the number of milliseconds on the CPU's system clock since the system was last booted, and then interpret this number as an IP address.^{70,71} SQL Slammer's ability to create a socket to connect two host nodes lies within the `Socket()` function. SQL Slammer utilizes this function to send packet clones to the PRNG incremented IP addresses over UDP Port 1434.⁷²

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe anomalous network traffic over UDP Port 1434, network traffic over TCP Port 179, UDP packet contents, and system behavior on local hosts.

A total of 97 observables were identified with the use of the [Native API technique \(T0834\)](#). This technique is important for investigation because it is the lowest-level means of execution to call to hardware, memory space, and process services for OS-based worms. This technique appears midway in the technique timeline and responding to it will limit all future scripted exploitation and propagation techniques on hosts with the [CVE-CAN-2002-0649](#) vulnerability. Terminating the chain of techniques at this point would limit SQL Slammer's outward propagation, rapid replication rate, and bandwidth saturation.

Of the 97 observables associated with this technique, 69 are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 25 artifacts could be generated by the Native API technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.6. EXPLOITATION FOR PRIVILEGE ESCALATION TECHNIQUE (T0890) FOR PRIVILEGE ESCALATION

Once a SQL Slammer clone has been sent to a vulnerable SQL Server 2000 host, arbitrary code is executed in memory, outside of the allocated fixed-size destination buffer reserved for the SQL Server Resolution Service Request.^{73,74,75} SQL Server 2000 runs in the security context chosen by the administrator at the time of installation: by default it runs as a domain user, but it also could run as a local SYSTEM user.⁷⁶ Therefore, depending on the SQL Server 2000 host that is infected, the arbitrary code will execute outside of the fixed-size destination buffer with domain level or SYSTEM level privileges.

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe anomalous network traffic over UDP Port 1434, network traffic over TCP Port 179, UDP packet contents, and system behavior on local hosts.

A total of 97 observables were identified with the use of the [Exploitation for Privilege Escalation technique \(T0890\)](#). This technique is important for investigation because it provides malicious, exploitative code the ability to run at an elevated privilege. This technique appears midway in the technique timeline and responding to it will limit all future scripted exploitation and propagation techniques on hosts with the [CVE-CAN-2002-0649](#) vulnerability. Terminating the chain of techniques at this point would limit SQL Slammer's outward propagation, rapid replication rate, and bandwidth saturation.

Of the 97 observables associated with this technique, 69 are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 16 artifacts could be generated by the Exploitation for Privilege Escalation technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.7. REMOTE SYSTEM DISCOVERY TECHNIQUE (T0846) FOR DISCOVERY

SQL Slammer rapidly scans, discovers, propagates to, and infects vulnerable hosts simultaneously, as packets are sent to a loop of pseudo-randomly generated IP addresses.^{77,78} Because single SQL Slammer worm packets are sent over UDP port 1434 and carry the potential to exploit vulnerable SQL Server 2000 hosts, the worm broadcasts its UDP scans without requiring responses from potential victim hosts, thus making it quicker and more effective.⁷⁹ This pseudo-random infinite IP loop for packet spraying, causing a flood of outbound UDP packets, is what led to the unintended consequence of performing denial of service attacks, with rapidly propagating packets saturating available bandwidth: SQL Slammer scans as fast as the infected host or network can transmit the packets.

SQL Slammer's scanning entails socket setups, followed by the start of an infinite loop of PRNG IP addresses (pseudo-randomly generated IP addresses) that create and send a packet copy of itself to the cycled PRNG IP addresses over UDP Port 1434.⁸⁰ In generating an infinite loop of IP addresses, SQL Slammer uses a linear congruent PRNG algorithm, where the GetTickCount() function seeds each PRNG scan cycle instance with the number of milliseconds on the CPU's system clock since the system was last booted, and then interprets this number as an IP address.^{81,82} After the PRNG scan cycle is seeded with GetTickCount(), the algorithm will infinitely increment and cycle according to the algorithm's increment value. The specific PRNG scan cycle utilized by SQL Slammer resulted in approximately 74,856 distinct IP addresses being infected across many domains and geographic locations. Though it did not limit SQL Slammer's spread much, several errors made by the author within the PRNG algorithm left out many IP addresses within the internet address space. Despite this, the algorithm generated public, private, broadcast, and multicast IP addresses.^{83,84,85}

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe anomalous network traffic over UDP Port 1434, network traffic over TCP Port 179, UDP packet contents, and system behavior on local hosts.

A total of 97 observables were identified with the use of the [Remote System Discovery technique \(T0846\)](#). This technique is important for investigation because it provides exploitative worms with a means to simultaneously scan, discover, propagate to, and infect vulnerable hosts with single UDP packets. This technique appears midway in the timeline and responding to it will limit all future scripted exploitation and propagation techniques on hosts with the [CVE-CAN-2002-0649](#) vulnerability. Terminating the chain of techniques at this point would limit SQL Slammer's bandwidth saturation.

Of the 97 observables associated with this technique, 69 are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 43 artifacts could be generated by the Remote System Discovery technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.8. EXPLOITATION OF REMOTE SERVICES TECHNIQUE (T0866) FOR LATERAL MOVEMENT

SQL Slammer's successive infections and outward propagation is due to a vulnerability in SQL Server 2000 Resolution Service. [CVE-CAN-2002-0649](#) allowed a deliberately designed packet, remotely sent to SQL Server 2000 Resolution Service over UDP Port 1434, to overwrite portions of system memory, granting remote execution of arbitrary, memory-injected code at an elevated privilege.⁸⁶ The remote execution of SQL Slammer's code triggers an infinite exploitation and propagation loop, which sends SQL Slammer packet clones to pseudo-randomly selected, PRNG IP addresses within a worm's scan cycle instance.^{87,88} The flood of requests is just meaningless network traffic until a single SQL Slammer packet reaches a vulnerable host. SQL Slammer's infinite exploitation and propagation loop routine will continue until the infected host is rebooted or crashes, as it remains solely in memory.⁸⁹

When SQL Server 2000 Resolution Service receives a SQL Slammer packet starting with a 0x04 byte, followed by many 0x01 bytes, the 0x01 bytes are inputted into the Sprintf() function and SQL Server Monitor thread generates a registry key to open with a long name of 'A' characters and causes a buffer overflow of the fixed-sized destination buffer.⁹⁰ This results in overwrites of key areas of memory with SQL Slammer's propagation code, while also gaining control over the SQL Server process.

IT Staff, IT Cybersecurity, and OT Cybersecurity personnel may have been able to observe anomalous network traffic over UDP Port 1434, network traffic over TCP Port 179, UDP packet contents, and system behavior on local hosts.

A total of 97 observables were identified with the use of the [Exploitation of Remote Services technique \(T0866\)](#). This technique is important for investigation because it is the mechanism by which an adversary can continually spread exploitative, self-propagating worms to vulnerable networks. This technique appears midway in the timeline and responding to it will limit all future scripted exploitation and propagation techniques on hosts with the [CVE-CAN-2002-0649](#) vulnerability. Terminating the chain of techniques at this point would limit SQL Slammer's outward propagation, rapid replication rate, and bandwidth saturation.

Of the 97 observables associated with this technique, 69 are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of 31 artifacts could be generated by the Exploitation of Remote Services technique
Technique Observers	IT Staff, IT Cybersecurity, OT Cybersecurity
Resources	Technique Detection References

3.9. LOSS OF AVAILABILITY TECHNIQUE (T0826) FOR IMPACT

At about 9:00 AM on 25 January, users on Davis-Besse's enterprise network noticed slow performance.⁹¹ By 4:00 PM, plant personnel noticed slow performance on the operations network. Then at 4:50 PM, the network congestion caused by SQL Slammer's intensive scanning crashed the SPDS. At 5:13 PM, the PPC crashed. Though operators had lost their digital monitoring systems, the SPDS and the PPC had redundant analog backups that could not be affected by the worm, and the plant control and safety functions were not affected.

The unintended payload of SQL Slammer is its scanning/propagation mechanism, as each successive SQL Slammer infection competes for bandwidth and scans as fast as the infected host or network transmits packets.⁹² Due to SQL Slammer's bandwidth saturation with its scanning, it causes major network congestion, slow performance, data overloads, increased latency, increased packet loss, denial of services, network outages, the inability of internal hosts to communicate with other internal hosts, and system crashes.^{93,94}

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe anomalous system behavior on local hosts.

A total of 22 observables were identified with the use of the [Loss of Availability technique \(T0826\)](#). This technique is important for investigation because it prevents organizations from delivering products or services. This technique appears late in the timeline and represents the triggering event. Responding to it has the potential to limit the extent of SQL Slammer's propagation within the internal network and saturation of available host and network bandwidth. Terminating the chain of techniques at this point would not have prevented Davis-Besse from being infected by the SQL Slammer worm.

All 22 observables associated with this technique are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of eight artifacts could be generated by the Loss of Availability technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

3.10. LOSS OF VIEW TECHNIQUE (T0829) FOR IMPACT

The SQL Slammer infection caused Davis-Besse's SPDS and PPC to crash.⁹⁵ The SPDS is the plant's computerized display panel that relays critical plant safety parameters to plant operators and the NRC Operations Center in real time. The SPDS monitors crucial safety measures such as coolant system sensor readouts, core temperature sensor readouts, and external radiation sensor readouts. Even though the plant was offline at the time of the SQL Slammer infection, many of the safety measures still require diligent monitoring. The PPC is a plant wide I/O information system for consolidated analog and digital data acquisition and transfer.^{96,97,98}

IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, and Management personnel may have been able to observe loss of view for various systems.

A total of 22 observables were identified with the use of the [Loss of View technique \(T0829\)](#). This technique is important for investigation because it causes a sustained loss of view of the digital monitoring systems. This technique appears late in the timeline and responding to it will not limit SQL Slammer's propagation within the internal network and saturation of available host and network bandwidth. Terminating the chain of techniques at this point would not prevented Davis-Besse from being infected by the SQL Slammer worm.

All 22 observables associated with this technique are assessed to be highly perceivable. They are italicized and marked † in [Appendix A](#).

CyOTE Capabilities for Technique Perception and Comprehension	
Artifacts	A total of four artifacts could be generated by the Loss of View technique
Technique Observers	IT Staff, IT Cybersecurity, OT Staff, OT Cybersecurity, Engineering, Support Staff, Management
Resources	Technique Detection References

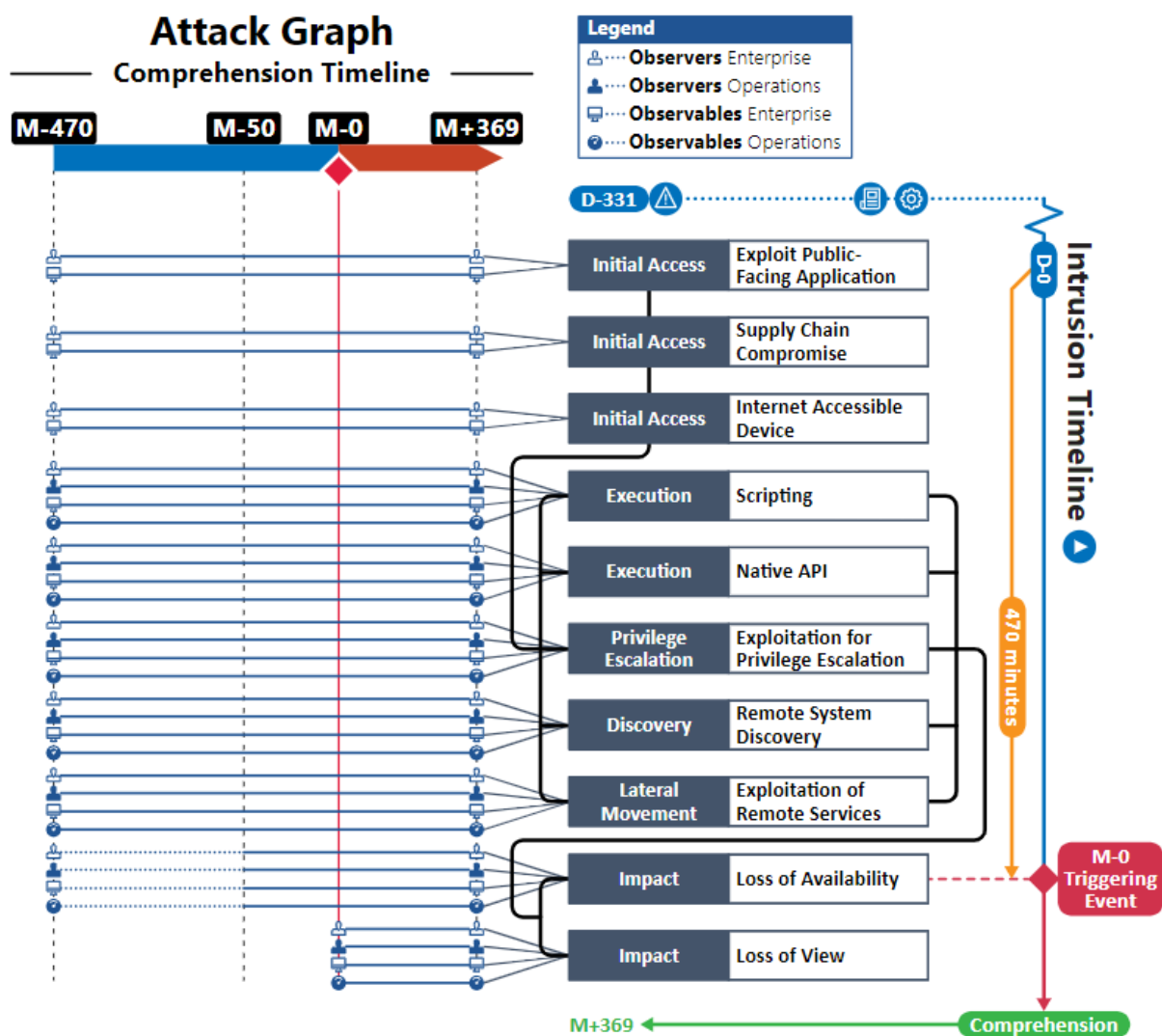


Figure 3. Attack Graph

APPENDIX A: OBSERVABLES LIBRARY

NOTE: Highly perceivable observables are italicized and marked †

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
Observable 1 †	<i>Presence of Vulnerability on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Critical Security Patch/Update</i>
Observable 2 †	<i>Presence of Vulnerability on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Microsoft Security Bulletin MS02-039 - Critical: Buffer Overruns in SQL Server 2000 Resolution Service Could Enable Code Execution (Q323875): CVE-CAN-2002-0649: Microsoft SQL Server 2000 Resolution Service Buffer Overflow Vulnerability</i>
Observable 3 †	<i>Presence of Vulnerability on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Carnegie Mellon University Software Engineering Institute CERT Coordination Center: CA-2002-22: Multiple Vulnerabilities in Microsoft SQL Server: Vulnerability Note VU#484891: CAN-2002-0649: Microsoft SQL Server 2000 Contains Stack Buffer Overflow in SQL Server Resolution Service</i>
Observable 4 †	<i>Presence of Vulnerability on Local Host: Microsoft Structured Query Language (SQL) Server 2000: CERT Advisory CA-2003-04 MS-SQL Server Worm</i>
Observable 5 †	<i>Presence of Vulnerability on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Microsoft Statement on the "Slammer" Worm Attack</i>
Observable 6 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 7 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 8 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 9 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 10 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-</i>

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
	<i>byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 11 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 12 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 13 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 14 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 15 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 16 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Public-Facing Hosts: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 17 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Inbound Sources: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 18 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 19 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "</i>

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
Observable 20 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 01 01 01 01 01 "</i>
Observable 21 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 22 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 23 †	<i>Anomalous Network Traffic: Between External Edge Routers and Internal Edge Routers: Over TCP Port 179: Border Gateway Protocol (BGP) Requests: Routing Update Messages: BGP Routing Table Updates</i>
Observable 24 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 25 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 26 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 27 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 28 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 29 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way</i>

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
	<i>Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 30 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 31 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 32 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 33 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 34 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 35 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 36 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 37 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 ; offset-0; depth-1"</i>
Observable 38 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 39 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server</i>

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
	2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01"
Observable 40 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
Observable 41 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte
Observable 42 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04
Observable 43 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00
Observable 44 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable
Observable 45 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out
Observable 46 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04
Observable 47 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
Observable 48 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 49 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 50 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 51 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 52 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 53 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 54 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Private Addresses</i>
Observable 55 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Broadcast IPs</i>
Observable 56 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Multicast IPs</i>
Observable 57 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 58 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server</i>

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
	2000: Over UDP Port 1434: UDP Packet Contents: Containing " 68 6F 75 6E 74 68 69 63 6B "
Observable 59 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 01 01 01 01 01 "
Observable 60 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
Observable 61 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte
Observable 62 †	Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable
Observable 63 †	Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out
Observable 64 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Service Process
Observable 65 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Resolution Service Process
Observable 66 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Monitor Process
Observable 67 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances
Observable 68 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways
Observable 69 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways
Observable 70 †	Anomalous System Behavior on Local Host: Saturation of Bandwidth: Microsoft Structured Query Language (SQL) Server 2000 Hosts
Observable 71 †	Anomalous System Behavior on Local Host: Saturation of Bandwidth: Local Area Network Gateways
Observable 72 †	Anomalous System Behavior on Local Host: Saturation of Bandwidth: Wide Area Network Gateways
Observable 73	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Sprintf()

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
Observable 74	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: LoadLibrary()
Observable 75	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetProcAddress()
Observable 76	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetTickCount()
Observable 77	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Socket()
Observable 78	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: ws2_32handle
Observable 79	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: kernel32handle
Observable 80	Anomalous System Behavior on Local Host: Anomalous Socket Setup: Microsoft Structured Query Language (SQL) Server 2000
Observable 81	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Generate Registry Key: Registry Key With Long Name
Observable 82	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server\AAAA\MSSQLServer\CurrentVersion
Observable 83	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server
Observable 84	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\MSSQLServer\CurrentVersion
Observable 85	Anomalous System Behavior on Local Host: Anomalous Allocation of Memory Space: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes
Observable 86	Anomalous System Behavior on Local Host: Anomalous Memory Writes: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes: Sprintf() Function Overwrites Destination Buffer With Too-Long String
Observable 87 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: Domain User</i>
Observable 88 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000:</i>

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
	<i>Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: SYSTEM User</i>
Observable 89	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Load the Address of a JMP ESP Instruction
Observable 90	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77F8313C
Observable 91	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77E89B18
Observable 92	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: ws2_32.dll
Observable 93	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: kernel32.dll
Observable 94	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EBX Register
Observable 95	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EAX Register
Observable 96	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77f8313c
Observable 97	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77e89b18
Observable 98	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77ea094c
Observable 99	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of

Observables Associated with Exploit Public-Facing Application Technique (T0819)	
	Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Disuse of Adding 1 in the Twos Complement Negation
Observable 100	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of ADD Instead of SUB to Compensate for the Resulting Negative Number
Observable 101	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: 25th and 26th Bits in the Scanned IP Addresses Remain Constant
Observable 102	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: Certain /16 Address Blocks not Included in Scanning Cycle: Last Two Bits of First Address Byte Never Change

Observables Associated with Supply Chain Compromise Technique (T0862)	
Observable 1 †	<i>Anomalous System Behavior on Local Host: Initial Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Host: Increase in Local Host Throughput Associated with Scanning: Over UDP Port 1434: Local Host with Consultant Transmission System 1 (T1) Connection: Server Running Consultant's Application Software</i>
Observable 2 †	<i>Anomalous System Behavior on Local Host: Initial Increase in System Resource Utilization: Local Area Network Gateway: Increase in Local Host Throughput Associated with Scanning: Over UDP Port 1434: Local Area Network with Consultant Transmission System 1 (T1) Connection: Local Corporate Network</i>
Observable 3 †	<i>Anomalous Installation of Data Communications Line: From External Corporate Entity to Local Environment: Transmission System 1 (T1) Line: Bridging Consultant Company Network to Local Corporate Network: Connected behind External-Facing Firewall: Used by Consultant</i>
Observable 4 †	<i>Presence of Unmonitored Network Traffic: From External Corporate Entity to Local Environment: Over UDP Port 1434: Not Routed Through Local Corporate Network Gateway: Bypasses External-Facing Firewall's Access Control Policy and Ruleset</i>

Observables Associated with Internet Accessible Device Technique (T0883)	
Observable 1 †	<i>Presence of Vulnerability on Local Host: Regulatory Body Provides Notification of Alert: Issuance of Anomalous Nuclear Regulatory Commission (NRC)</i>

Observables Associated with Internet Accessible Device Technique (T0883)	
	<i>Security Order: Alert for External Network Connections Bypassing Network Boundary Protections</i>
Observable 2 †	<i>Anomalous System Behavior on Local Host: Initial Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Host: Increase in Local Host Throughput Associated with Scanning: Over UDP Port 1434: Local Host with Consultant Transmission System 1 (T1) Connection: Server Running Consultant's Application Software</i>
Observable 3 †	<i>Anomalous System Behavior on Local Host: Initial Increase in System Resource Utilization: Local Area Network Gateway: Increase in Local Host Throughput Associated with Scanning: Over UDP Port 1434: Local Area Network with Consultant Transmission System 1 (T1) Connection: Local Corporate Network</i>
Observable 4 †	<i>Anomalous Installation of Data Communications Line: From External Corporate Entity to Local Environment: Transmission System 1 (T1) Line: Bridging Consultant Company Network to Local Corporate Network: Connected behind External-Facing Firewall: Used by Consultant</i>
Observable 5 †	<i>Presence of Unmonitored Network Traffic: From External Corporate Entity to Local Environment: Over UDP Port 1434: Not Routed Through Local Corporate Network Gateway: Bypasses External-Facing Firewall's Access Control Policy and Ruleset</i>

Observables Associated with Scripting Technique (T0853)	
Observable 1 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 2 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 3 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 4 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>

Observables Associated with Scripting Technique (T0853)	
Observable 5 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 6 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 7 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 8 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 9 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 10 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 11 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Public-Facing Hosts: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 12 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Inbound Sources: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 13 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 \; offset-0; depth-1"</i>
Observable 14 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000:</i>

Observables Associated with Scripting Technique (T0853)	
	Over UDP Port 1434: UDP Packet Contents: Containing " 68 6F 75 6E 74 68 69 63 6B "
Observable 15 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 01 01 01 01 01 "
Observable 16 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
Observable 17 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte
Observable 18 †	Anomalous Network Traffic: Between External Edge Routers and Internal Edge Routers: Over TCP Port 179: Border Gateway Protocol (BGP) Requests: Routing Update Messages: BGP Routing Table Updates
Observable 19 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04
Observable 20 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00
Observable 21 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable
Observable 22 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out
Observable 23 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04

Observables Associated with Scripting Technique (T0853)	
Observable 24 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 25 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 26 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 27 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 28 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 29 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 30 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 31 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 32 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 ; offset-0; depth-1"</i>
Observable 33 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 68 6F 75 6E 74 68 69 63 6B "</i>

Observables Associated with Scripting Technique (T0853)	
Observable 34 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01"</i>
Observable 35 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 36 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 37 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 38 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 39 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 40 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 41 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 42 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-</i>

Observables Associated with Scripting Technique (T0853)	
	<i>byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 43 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 44 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 45 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 46 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 47 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 48 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 49 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Private Addresses</i>
Observable 50 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Broadcast IPs</i>
Observable 51 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Multicast IPs</i>
Observable 52 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>

Observables Associated with Scripting Technique (T0853)	
Observable 53 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B \"</i>
Observable 54 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 \"</i>
Observable 55 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 56 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 57 †	<i>Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 58 †	<i>Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 59 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Service Process</i>
Observable 60 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Resolution Service Process</i>
Observable 61 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Monitor Process</i>
Observable 62 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 63 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways</i>
Observable 64 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways</i>
Observable 65 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Microsoft Structured Query Language (SQL) Server 2000 Hosts</i>
Observable 66 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Local Area Network Gateways</i>
Observable 67 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Wide Area Network Gateways</i>

Observables Associated with Scripting Technique (T0853)	
Observable 68	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: <code>Sprintf()</code>
Observable 69	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: <code>LoadLibrary()</code>
Observable 70	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: <code>GetProcAddress()</code>
Observable 71	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: <code>GetTickCount()</code>
Observable 72	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: <code>Socket()</code>
Observable 73	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: <code>ws2_32handle</code>
Observable 74	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: <code>kernel32handle</code>
Observable 75	Anomalous System Behavior on Local Host: Anomalous Socket Setup: Microsoft Structured Query Language (SQL) Server 2000
Observable 76	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Generate Registry Key: Registry Key With Long Name
Observable 77	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: <code>HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server\AAAA\MSSQLServer\CurrentVersion</code>
Observable 78	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: <code>HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server</code>
Observable 79	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: <code>HKEY_LOCAL_MACHINE\Software\Microsoft\MSSQLServer\CurrentVersion</code>
Observable 80	Anomalous System Behavior on Local Host: Anomalous Allocation of Memory Space: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes
Observable 81	Anomalous System Behavior on Local Host: Anomalous Memory Writes: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes: <code>Sprintf()</code> Function Overwrites Destination Buffer With Too-Long String
Observable 82 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000:</i>

Observables Associated with Scripting Technique (T0853)	
	<i>Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: Domain User</i>
Observable 83 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: SYSTEM User</i>
Observable 84	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Load the Address of a JMP ESP Instruction
Observable 85	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77F8313C
Observable 86	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77E89B18
Observable 87	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: ws2_32.dll
Observable 88	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: kernel32.dll
Observable 89	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EBX Register
Observable 90	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EAX Register
Observable 91	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77f8313c
Observable 92	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77e89b18
Observable 93	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for

Observables Associated with Scripting Technique (T0853)	
	Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77ea094c
Observable 94	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Disuse of Adding 1 in the Twos Complement Negation
Observable 95	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of ADD Instead of SUB to Compensate for the Resulting Negative Number
Observable 96	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: 25th and 26th Bits in the Scanned IP Addresses Remain Constant
Observable 97	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: Certain /16 Address Blocks not Included in Scanning Cycle: Last Two Bits of First Address Byte Never Change

Observables Associated with Native API Technique (T0834)	
Observable 1 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 2 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 3 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>

Observables Associated with Native API Technique (T0834)	
Observable 4 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 5 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 6 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 7 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 8 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 9 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 10 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 11 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Public-Facing Hosts: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 12 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000:</i>

Observables Associated with Native API Technique (T0834)	
	Over UDP Port 1434: Numerous New Inbound Sources: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses
Observable 13 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 ; offset-0; depth-1"
Observable 14 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 68 6F 75 6E 74 68 69 63 6B "
Observable 15 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 01 01 01 01 01 "
Observable 16 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
Observable 17 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte
Observable 18 †	Anomalous Network Traffic: Between External Edge Routers and Internal Edge Routers: Over TCP Port 179: Border Gateway Protocol (BGP) Requests: Routing Update Messages: BGP Routing Table Updates
Observable 19 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04
Observable 20 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00
Observable 21 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable
Observable 22 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434

Observables Associated with Native API Technique (T0834)	
	<i>Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 23 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 24 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 25 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 26 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 27 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 28 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 29 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 30 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 31 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>

Observables Associated with Native API Technique (T0834)	
Observable 32 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 33 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 34 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "</i>
Observable 35 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 36 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 37 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 38 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 39 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 40 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 41 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way</i>

Observables Associated with Native API Technique (T0834)	
	<i>Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 42 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 43 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 44 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 45 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 46 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 47 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 48 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 49 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Private Addresses</i>
Observable 50 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Broadcast IPs</i>

Observables Associated with Native API Technique (T0834)	
Observable 51 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Multicast IPs</i>
Observable 52 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 53 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 54 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "</i>
Observable 55 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 56 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 57 †	<i>Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 58 †	<i>Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 59 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Service Process</i>
Observable 60 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Resolution Service Process</i>
Observable 61 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Monitor Process</i>
Observable 62 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 63 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways</i>
Observable 64 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways</i>

Observables Associated with Native API Technique (T0834)	
Observable 65 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Microsoft Structured Query Language (SQL) Server 2000 Hosts</i>
Observable 66 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Local Area Network Gateways</i>
Observable 67 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Wide Area Network Gateways</i>
Observable 68	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Sprintf()
Observable 69	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: LoadLibrary()
Observable 70	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetProcAddress()
Observable 71	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetTickCount()
Observable 72	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Socket()
Observable 73	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: ws2_32handle
Observable 74	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: kernel32handle
Observable 75	Anomalous System Behavior on Local Host: Anomalous Socket Setup: Microsoft Structured Query Language (SQL) Server 2000
Observable 76	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Generate Registry Key: Registry Key With Long Name
Observable 77	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server\AAAA\MSSQLServer\CurrentVersion
Observable 78	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server
Observable 79	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\MSSQLServer\CurrentVersion
Observable 80	Anomalous System Behavior on Local Host: Anomalous Allocation of Memory Space: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes

Observables Associated with Native API Technique (T0834)	
Observable 81	Anomalous System Behavior on Local Host: Anomalous Memory Writes: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes: Sprintf() Function Overwrites Destination Buffer With Too-Long String
Observable 82 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: Domain User</i>
Observable 83 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: SYSTEM User</i>
Observable 84	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Load the Address of a JMP ESP Instruction
Observable 85	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77F8313C
Observable 86	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77E89B18
Observable 87	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: ws2_32.dll
Observable 88	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: kernel32.dll
Observable 89	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EBX Register
Observable 90	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EAX Register
Observable 91	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77f8313c
Observable 92	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for

Observables Associated with Native API Technique (T0834)	
	Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77e89b18
Observable 93	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77ea094c
Observable 94	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Disuse of Adding 1 in the Twos Complement Negation
Observable 95	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of ADD Instead of SUB to Compensate for the Resulting Negative Number
Observable 96	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: 25th and 26th Bits in the Scanned IP Addresses Remain Constant
Observable 97	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: Certain /16 Address Blocks not Included in Scanning Cycle: Last Two Bits of First Address Byte Never Change

Observables Associated with Exploitation for Privilege Escalation Technique (T0890)	
Observable 1 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 2 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>

Observables Associated with Exploitation for Privilege Escalation Technique (T0890)	
Observable 3 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 4 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 5 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 6 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 7 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 8 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 9 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 10 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 11 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000:</i>

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	Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Public-Facing Hosts: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances
Observable 12 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Inbound Sources: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses
Observable 13 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"
Observable 14 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "
Observable 15 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "
Observable 16 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
Observable 17 †	Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte
Observable 18 †	Anomalous Network Traffic: Between External Edge Routers and Internal Edge Routers: Over TCP Port 179: Border Gateway Protocol (BGP) Requests: Routing Update Messages: BGP Routing Table Updates
Observable 19 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04
Observable 20 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00
Observable 21 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable

Observables Associated with Exploitation for Privilege Escalation Technique (T0890)	
Observable 22 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 23 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 24 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 25 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 26 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 27 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 28 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 29 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 30 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server</i>

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	2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways
Observable 31 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses
Observable 32 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"
Observable 33 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "
Observable 34 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "
Observable 35 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
Observable 36 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte
Observable 37 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04
Observable 38 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00
Observable 39 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable
Observable 40 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-

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	<i>byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 41 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 42 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 43 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 44 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 45 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 46 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 47 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 48 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 49 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server</i>

Observables Associated with Exploitation for Privilege Escalation Technique (T0890)	
	2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Private Addresses
Observable 50 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Broadcast IPs
Observable 51 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Multicast IPs
Observable 52 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"
Observable 53 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "
Observable 54 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "
Observable 55 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
Observable 56 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte
Observable 57 †	Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable
Observable 58 †	Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out
Observable 59 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Service Process
Observable 60 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Resolution Service Process
Observable 61 †	Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Monitor Process

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Observable 62 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 63 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways</i>
Observable 64 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways</i>
Observable 65 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Microsoft Structured Query Language (SQL) Server 2000 Hosts</i>
Observable 66 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Local Area Network Gateways</i>
Observable 67 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Wide Area Network Gateways</i>
Observable 68	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Sprintf()
Observable 69	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: LoadLibrary()
Observable 70	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetProcAddress()
Observable 71	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetTickCount()
Observable 72	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Socket()
Observable 73	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: ws2_32handle
Observable 74	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: kernel32handle
Observable 75	Anomalous System Behavior on Local Host: Anomalous Socket Setup: Microsoft Structured Query Language (SQL) Server 2000
Observable 76	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Generate Registry Key: Registry Key With Long Name
Observable 77	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server\AAAA\MSSQLServer\CurrentVersion
Observable 78	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server

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Observable 79	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\MSSQLServer\CurrentVersion
Observable 80	Anomalous System Behavior on Local Host: Anomalous Allocation of Memory Space: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes
Observable 81	Anomalous System Behavior on Local Host: Anomalous Memory Writes: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes: Sprintf() Function Overwrites Destination Buffer With Too-Long String
Observable 82 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: Domain User</i>
Observable 83 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: SYSTEM User</i>
Observable 84	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Load the Address of a JMP ESP Instruction
Observable 85	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77F8313C
Observable 86	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77E89B18
Observable 87	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: ws2_32.dll
Observable 88	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: kernel32.dll
Observable 89	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EBX Register
Observable 90	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EAX Register
Observable 91	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of

Observables Associated with Exploitation for Privilege Escalation Technique (T0890)	
	Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77f8313c
Observable 92	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77e89b18
Observable 93	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77ea094c
Observable 94	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Disuse of Adding 1 in the Twos Complement Negation
Observable 95	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of ADD Instead of SUB to Compensate for the Resulting Negative Number
Observable 96	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: 25th and 26th Bits in the Scanned IP Addresses Remain Constant
Observable 97	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: Certain /16 Address Blocks not Included in Scanning Cycle: Last Two Bits of First Address Byte Never Change

Observables Associated with Remote System Discovery Technique (T0846)	
Observable 1 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>

Observables Associated with Remote System Discovery Technique (T0846)	
Observable 2 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 3 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 4 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 5 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 6 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 7 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 8 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 9 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>

Observables Associated with Remote System Discovery Technique (T0846)	
Observable 10 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 11 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Public-Facing Hosts: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 12 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Inbound Sources: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 13 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 14 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 15 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "</i>
Observable 16 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 17 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 18 †	<i>Anomalous Network Traffic: Between External Edge Routers and Internal Edge Routers: Over TCP Port 179: Border Gateway Protocol (BGP) Requests: Routing Update Messages: BGP Routing Table Updates</i>
Observable 19 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 20 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>

Observables Associated with Remote System Discovery Technique (T0846)	
Observable 21 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 22 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 23 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 24 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 25 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 26 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 27 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 28 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>

Observables Associated with Remote System Discovery Technique (T0846)	
Observable 29 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 30 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 31 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 32 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 33 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 34 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "</i>
Observable 35 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 36 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 37 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 38 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 39 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434</i>

Observables Associated with Remote System Discovery Technique (T0846)	
	<i>Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 40 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 41 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 42 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 43 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 44 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 45 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 46 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 47 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>

Observables Associated with Remote System Discovery Technique (T0846)	
Observable 48 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 49 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Private Addresses</i>
Observable 50 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Broadcast IPs</i>
Observable 51 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Multicast IPs</i>
Observable 52 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 53 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 54 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "</i>
Observable 55 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 56 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 57 †	<i>Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 58 †	<i>Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 59 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Service Process</i>

Observables Associated with Remote System Discovery Technique (T0846)	
Observable 60 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Resolution Service Process</i>
Observable 61 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Monitor Process</i>
Observable 62 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 63 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways</i>
Observable 64 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways</i>
Observable 65 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Microsoft Structured Query Language (SQL) Server 2000 Hosts</i>
Observable 66 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Local Area Network Gateways</i>
Observable 67 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Wide Area Network Gateways</i>
Observable 68	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Sprintf()
Observable 69	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: LoadLibrary()
Observable 70	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetProcAddress()
Observable 71	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: GetTickCount()
Observable 72	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Socket()
Observable 73	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: ws2_32handle
Observable 74	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: kernel32handle
Observable 75	Anomalous System Behavior on Local Host: Anomalous Socket Setup: Microsoft Structured Query Language (SQL) Server 2000
Observable 76	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Generate Registry Key: Registry Key With Long Name
Observable 77	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open

Observables Associated with Remote System Discovery Technique (T0846)	
	Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server\AAAA\MSSQLServer\CurrentVersion
Observable 78	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server
Observable 79	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\MSSQLServer\CurrentVersion
Observable 80	Anomalous System Behavior on Local Host: Anomalous Allocation of Memory Space: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes
Observable 81	Anomalous System Behavior on Local Host: Anomalous Memory Writes: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes: Sprintf() Function Overwrites Destination Buffer With Too-Long String
Observable 82 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: Domain User</i>
Observable 83 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: SYSTEM User</i>
Observable 84	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Load the Address of a JMP ESP Instruction
Observable 85	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77F8313C
Observable 86	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77E89B18
Observable 87	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: ws2_32.dll
Observable 88	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: kernel32.dll
Observable 89	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EBX Register
Observable 90	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000:

Observables Associated with Remote System Discovery Technique (T0846)	
	Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EAX Register
Observable 91	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77f8313c
Observable 92	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77e89b18
Observable 93	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of OR Instead of XOR to Clear the Value of EBX: 0x77ea094c
Observable 94	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Disuse of Adding 1 in the Twos Complement Negation
Observable 95	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: Use of ADD Instead of SUB to Compensate for the Resulting Negative Number
Observable 96	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: 25th and 26th Bits in the Scanned IP Addresses Remain Constant
Observable 97	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$: 74,856 Distinctly Infected IP Addresses: Certain /16 Address Blocks not Included in Scanning Cycle: Last Two Bits of First Address Byte Never Change

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Observable 1 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 2 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 3 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 4 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 5 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 6 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 7 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 8 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434</i>

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	<i>Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 9 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 10 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 11 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Public-Facing Hosts: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 12 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Inbound Sources: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 13 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 ; offset-0; depth-1"</i>
Observable 14 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 15 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing " 04 01 01 01 01 01 "</i>
Observable 16 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 17 †	<i>Anomalous Network Traffic: From Pseudo-random External Hosts to Internal Application Servers: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 18 †	<i>Anomalous Network Traffic: Between External Edge Routers and Internal Edge Routers: Over TCP Port 179: Border Gateway Protocol (BGP) Requests: Routing Update Messages: BGP Routing Table Updates</i>
Observable 19 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>

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Observable 20 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 21 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 22 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 23 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 24 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 25 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 26 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 27 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>

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Observable 28 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 29 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways</i>
Observable 30 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways</i>
Observable 31 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Public Addresses</i>
Observable 32 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"</i>
Observable 33 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "</i>
Observable 34 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "</i>
Observable 35 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte</i>
Observable 36 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random External Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Absence of End Flag Byte: 0x00 Flag Byte</i>
Observable 37 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 38 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-</i>

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	<i>byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 39 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 40 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 376-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 41 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: With First Byte 0x04</i>
Observable 42 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Too-Long UDP Request for SQL Server Monitor: Without End Byte 0x00</i>
Observable 43 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable</i>
Observable 44 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: SQL Server Resolution Service Requests: One-Way Single Packets With No Associated Response: Pseudo-random Flood of 404-byte UDP Packet Clones With Exploit Code: Sent to Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 45 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Microsoft SQL Server 2000 Hosts</i>
Observable 46 †	<i>Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With</i>

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	Scanning: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances
Observable 47 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Local Area Network Gateways
Observable 48 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Increase in Network Traffic Associated With Scanning: Wide Area Network Gateways
Observable 49 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Unicast IPs: Private Addresses
Observable 50 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Broadcast IPs
Observable 51 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: Numerous New Outbound Destinations: 74,856 Distinctly Infected IP Addresses: Multicast IPs
Observable 52 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 ; offset-0; depth-1"
Observable 53 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 68 6F 75 6E 74 68 69 63 6B "
Observable 54 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Containing "\ 04 01 01 01 01 01 "
Observable 55 †	Anomalous Network Traffic: From Internal Application Servers to Pseudo-random Internal Hosts: Microsoft Structured Query Language (SQL) Server 2000: Over UDP Port 1434: UDP Packet Contents: Presence of First Flag Byte: 0x04 Flag Byte
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Observable 57 †	Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Destination Unreachable

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Observable 58 †	<i>Anomalous System Behavior on Local Host: Hosts With Port 1434 Closed: Packet Drops From Unavailable Service: Internet Control Message Protocol (ICMP) Error Code Response: Request Timed Out</i>
Observable 59 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Service Process</i>
Observable 60 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: SQL Server Resolution Service Process</i>
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Observable 62 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances</i>
Observable 63 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways</i>
Observable 64 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways</i>
Observable 65 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Microsoft Structured Query Language (SQL) Server 2000 Hosts</i>
Observable 66 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Local Area Network Gateways</i>
Observable 67 †	<i>Anomalous System Behavior on Local Host: Saturation of Bandwidth: Wide Area Network Gateways</i>
Observable 68	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: Sprintf()
Observable 69	Anomalous System Behavior on Local Host: Anomalous Call to Windows API on Local Host: Microsoft Structured Query Language (SQL) Server 2000: LoadLibrary()
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Observable 73	Anomalous System Behavior on Local Host: Anomalous Usage of Get Handles: Microsoft Structured Query Language (SQL) Server 2000: ws2_32handle
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Observable 75	Anomalous System Behavior on Local Host: Anomalous Socket Setup: Microsoft Structured Query Language (SQL) Server 2000
Observable 76	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Generate Registry Key: Registry Key With Long Name
Observable 77	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server\AAAA\MSSQLServer\CurrentVersion
Observable 78	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\Microsoft SQL Server
Observable 79	Anomalous System Behavior on Local Host: Anomalous Usage of Structured Query Language (SQL) Monitor Thread: Microsoft SQL Server 2000: Open Registry Key: HKEY_LOCAL_MACHINE\Software\Microsoft\MSSQLServer\CurrentVersion
Observable 80	Anomalous System Behavior on Local Host: Anomalous Allocation of Memory Space: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes
Observable 81	Anomalous System Behavior on Local Host: Anomalous Memory Writes: Microsoft Structured Query Language (SQL) Server 2000: Fixed Sized Destination Buffer on Stack Frame: 128 Bytes: Sprintf() Function Overwrites Destination Buffer With Too-Long String
Observable 82 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: Domain User</i>
Observable 83 †	<i>Anomalous System Behavior on Local Host: Anomalous Remote Execution of Arbitrary Code: Microsoft Structured Query Language (SQL) Server 2000: Sprintf() Function Passes Too-Long SQL Server UDP Request String: Using SQL Server Service Permissions: SQL Server Service Account: SYSTEM User</i>
Observable 84	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Load the Address of a JMP ESP Instruction
Observable 85	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77F8313C
Observable 86	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: SQLsort.dll: Call to SqlSort Import Address Table (IAT): Uses Salt Value as Increment Pointer: 0x77E89B18
Observable 87	Anomalous System Behavior on Local Host: Presence of Anomalous Library Imports: Microsoft Structured Query Language (SQL) Server 2000: ws2_32.dll
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Observable 89	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EBX Register
Observable 90	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: PRNG Seed: GetTickCount(): EAX Register
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Observable 97	Anomalous System Behavior on Local Host: Presence of Anomalous Pseudo-random Scan Cycle: Microsoft Structured Query Language (SQL) Server 2000: Scanning of Pseudo-random IP Addresses: Over UDP Port 1434: Usage of

Observables Associated with Exploitation of Remote Services Technique (T0866)

	Linear Congruent Pseudo-random Number Generation (PRNG) Algorithm for Infinite IP Address Scan Cycle: $\text{mod } 2^{32} \text{ addr}' = (\text{addr} * 214013 + (-2531012 \text{ xor } \text{EBX}))$; 74,856 Distinctly Infected IP Addresses: Certain /16 Address Blocks not Included in Scanning Cycle: Last Two Bits of First Address Byte Never Change
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Observables Associated with Loss of Availability Technique (T0826)

Observable 1 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput: Bandwidth Cap</i>
Observable 2 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 3 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 4 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput</i>
Observable 5 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 6 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 7 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput: Bandwidth Cap</i>
Observable 8 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 9 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 10 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput: Bandwidth Cap</i>
Observable 11 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways: Increase in Local Host Throughput</i>

Observables Associated with Loss of Availability Technique (T0826)	
	<i>Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 12 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 13 †	<i>Anomalous System Behavior on Local Host: Anomalous Slow Performance</i>
Observable 14 †	<i>Anomalous System Behavior on Local Host: Anomalous Data Overload</i>
Observable 15 †	<i>Anomalous System Behavior on Local Host: Anomalous Bandwidth Saturation</i>
Observable 16 †	<i>Anomalous System Behavior on Local Host: Anomalous Increase in Latency</i>
Observable 17 †	<i>Anomalous System Behavior on Local Host: Anomalous Increase in Packet Loss</i>
Observable 18 †	<i>Anomalous System Behavior on Local Host: Anomalous Denial of Service</i>
Observable 19 †	<i>Anomalous System Behavior on Local Host: Anomalous Network Outage</i>
Observable 20 †	<i>Anomalous System Behavior on Local Host: Anomalous Inability of Internal Hosts to Communicate With Other Internal Hosts</i>
Observable 21 †	<i>Anomalous System Behavior on Local Host: Anomalous System Crashes: Safety Parameter Display System (SPDS)</i>
Observable 22 †	<i>Anomalous System Behavior on Local Host: Anomalous System Crashes: Plant Process Computer (PPC)</i>

Observables Associated with Loss of View Technique (T0829)	
Observable 1 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput: Bandwidth Cap</i>
Observable 2 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 3 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Microsoft Structured Query Language (SQL) Server 2000 Hosts: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 4 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput</i>
Observable 5 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan Cycle Instances: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 6 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Hosts With IP Addresses Within Pseudo-random IP Generation Scan</i>

Observables Associated with Loss of View Technique (T0829)	
	<i>Cycle Instances: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 7 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput: Bandwidth Cap</i>
Observable 8 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 9 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Local Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 10 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Network Interface/Adapter Throughput: Bandwidth Cap</i>
Observable 11 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Central Processing Unit (CPU) Utilization</i>
Observable 12 †	<i>Anomalous System Behavior on Local Host: Increase in System Resource Utilization: Wide Area Network Gateways: Increase in Local Host Throughput Associated With Scanning: Over UDP Port 1434: Increase in Memory Usage</i>
Observable 13 †	<i>Anomalous System Behavior on Local Host: Anomalous Slow Performance</i>
Observable 14 †	<i>Anomalous System Behavior on Local Host: Anomalous Data Overload</i>
Observable 15 †	<i>Anomalous System Behavior on Local Host: Anomalous Bandwidth Saturation</i>
Observable 16 †	<i>Anomalous System Behavior on Local Host: Anomalous Increase in Latency</i>
Observable 17 †	<i>Anomalous System Behavior on Local Host: Anomalous Increase in Packet Loss</i>
Observable 18 †	<i>Anomalous System Behavior on Local Host: Anomalous Denial of Service</i>
Observable 19 †	<i>Anomalous System Behavior on Local Host: Anomalous Network Outage</i>
Observable 20 †	<i>Anomalous System Behavior on Local Host: Anomalous Inability of Internal Hosts to Communicate With Other Internal Hosts</i>
Observable 21 †	<i>Anomalous System Behavior on Local Host: Anomalous System Crashes: Safety Parameter Display System (SPDS)</i>
Observable 22 †	<i>Anomalous System Behavior on Local Host: Anomalous System Crashes: Plant Process Computer (PPC)</i>

APPENDIX B: ARTIFACTS LIBRARY

Artifacts Associated with Exploit Public-Facing Application Technique (T0819)	
Artifact 1	Logon Security Event
Artifact 2	Logon Timestamp
Artifact 3	Process Failure
Artifact 4	Process State Change
Artifact 5	Operational Data Modification
Artifact 6	Operational Data Corruption
Artifact 7	Open Platform Communication (OPC) Communication (COM) Objects
Artifact 8	Remote Connections
Artifact 9	External Network Connections
Artifact 10	Logon Event
Artifact 11	Prefetch
Artifact 12	Logon Event
Artifact 13	Administrator Logon
Artifact 14	External Network Connections
Artifact 15	Remote Connections
Artifact 16	Ransom Note
Artifact 17	Logon Timestamp After Hours
Artifact 18	MAC Address
Artifact 19	IP Address
Artifact 20	Process Ending
Artifact 21	HTTP Traffic Port
Artifact 22	External Industrial Protocol Connections
Artifact 23	Web Server Log
Artifact 24	Virtual Network Computing (VNC) Traffic Port
Artifact 25	Secure Shell (SSH) Traffic Port
Artifact 26	Logon Security Event
Artifact 27	Telnet Traffic
Artifact 28	Increase Number of Logon Attempts
Artifact 29	Trivial File Transfer Protocol (TFTP) Port
Artifact 30	File Transfer Protocol (FTP) Port
Artifact 31	Application Failure
Artifact 32	HTTPS Port
Artifact 33	User Account

Artifacts Associated with Exploit Public-Facing Application Technique (T0819)	
Artifact 34	Web Proxy Logs
Artifact 35	Application Log
Artifact 36	Process Creation
Artifact 37	Process Ending
Artifact 38	Source IP Address
Artifact 39	MAC Address
Artifact 40	Firewall Logs
Artifact 41	Transport Layer Security (TLP) Certificate
Artifact 42	.lnk Files
Artifact 43	File Transfer Protocol Secure (FTPS) Port
Artifact 44	Logon Alert for Default Password
Artifact 45	Process Creation
Artifact 46	Vendor Jump Host Logon
Artifact 47	Configuration Alert for Default Password
Artifact 48	Remote Connections
Artifact 49	Remote Desktop Protocol (RDP) Traffic Port
Artifact 50	VNC Traffic Port
Artifact 51	SSH Traffic Port
Artifact 52	Telnet Traffic
Artifact 53	HTTP Traffic
Artifact 54	Application Log
Artifact 55	RDP Traffic Port

Artifacts Associated with Supply Chain Compromise Technique (T0862)	
Artifact 1	MAC Address
Artifact 2	Link Layer Discovery Protocol (LLDP) Requests
Artifact 3	Usage of Vendor Maintenance Account
Artifact 4	Usage of Default Account
Artifact 5	Static Source IP Address
Artifact 6	Ping Echo Port
Artifact 7	HTTP Port
Artifact 8	Simple Network Mail Protocol (SNMP) Port
Artifact 9	Server Message Block (SMB) Port
Artifact 10	Network Discover Protocols

Artifacts Associated with Supply Chain Compromise Technique (T0862)	
Artifact 11	Domain Name
Artifact 12	Source IP Address
Artifact 13	Mismatched Software Hashes
Artifact 14	Domain Name System (DNS) Queries Traffic Port
Artifact 15	Inaccurate Delivery Based on Design Documents
Artifact 16	Destination IP Address
Artifact 17	Physical Defects to Hardware
Artifact 18	Factory Acceptance Test Failure
Artifact 19	Inconsistencies in Hardware Bill of Materials (HBOM)
Artifact 20	Inconsistencies in Software Bill of Materials (SBOM)
Artifact 21	Hardware Serial Number Missing
Artifact 22	Unscheduled Firmware Updates
Artifact 23	Domain Registrant Data
Artifact 24	Hardware Failed Site Acceptance Test
Artifact 25	Hardware Tampering Evidence
Artifact 26	Device Incompatibility Issues
Artifact 27	Device Failures
Artifact 28	Additional Hardware Inserted on Devices
Artifact 29	Domain Autonomous System Number
Artifact 30	Domain IP Resolution
Artifact 31	Manipulation of Signature on Digital Certifications

Artifacts Associated with Internet Accessible Device Technique (T0883)	
Artifact 1	Host Registry Entries
Artifact 2	HTTPS Traffic
Artifact 3	Suspicious Connections in Proxy Logs
Artifact 4	Timestamps
Artifact 5	Virtual Private Network (VPN) Logoff Events
Artifact 6	Suspicious Connections in Firewall Logs
Artifact 7	VPN Logon Events
Artifact 8	Systems Applications and Protocols (SAP) Traffic
Artifact 9	Host Registry Entries HKEY_LOCAL_MACHINE\SYSTEM
Artifact 10	SQL Traffic
Artifact 11	Host Information in External Data Store or Website (SHODAN)

Artifacts Associated with Internet Accessible Device Technique (T0883)	
Artifact 12	HTTP 80
Artifact 13	VNC Traffic Port 5800
Artifact 14	Dialog Boxes Opened on Human Machine Interface (HMI)
Artifact 15	Application Authentication Events
Artifact 16	Internet Address in Memory Socket Data
Artifact 17	Remote Logins in OS Logs (Windows Event)
Artifact 18	Operational Database Connection to External Addresses
Artifact 19	Industrial Traffic from Internet Address
Artifact 20	Standard Traffic from Internet Address
Artifact 21	Internet Address in Application Logs
Artifact 22	Internet Address in OS Logs
Artifact 23	Internet Address in Command Line Record Data (netstat)

Artifacts Associated with Scripting Technique (T0853)	
Artifact 1	Startup Menu Modification
Artifact 2	OS Service Installation
Artifact 3	Registry Modifications
Artifact 4	Network Services Created
Artifact 5	External Network Connections
Artifact 6	Prefetch Files Created
Artifact 7	Executable Files
Artifact 8	System Processes Created
Artifact 9	OS Timeline Event
Artifact 10	System Event Log Creation
Artifact 11	Files Dropped into Directory
Artifact 12	Windows API Event Log

Artifacts Associated with Native API Technique (T0834)	
Artifact 1	Alert Generated
Artifact 2	System Resource Usage Management Changes
Artifact 3	.dll Modifications
Artifact 4	Imports Hash Changed
Artifact 5	Files Created
Artifact 6	Processes Initiated

Artifacts Associated with Native API Technique (T0834)	
Artifact 7	Services Initiated
Artifact 8	SYSMON Events Created
Artifact 9	Performance Degradation
Artifact 10	Blue Screen
Artifact 11	Configuration Change
Artifact 12	Command Execution
Artifact 13	Industrial Protocol Command Packet
Artifact 14	Host Device Failure
Artifact 15	Industrial Network Traffic
Artifact 16	Device Reads
Artifact 17	Device I/O Image Table Manipulated
Artifact 18	Device Failure
Artifact 19	Systems Calls
Artifact 20	Device Performance Degradation
Artifact 21	Device Memory Modification
Artifact 22	Device Alarm
Artifact 23	Device Live Data Changes
Artifact 24	Alter Process Logic
Artifact 25	Memory Corruption

Artifacts Associated with Exploitation for Privilege Escalation Technique (T0890)	
Artifact 1	SYSMON Event 8 CREATEREMOTETHREAD Process Injection Detected
Artifact 2	Unexpected Process Crash
Artifact 3	Network Traffic Associated with Privilege Escalation Vulnerabilities (CVE-2014-4076 Sent a Specially Crafted TCP Packet to \\.\ TCP Device Through DEVICEIOCONTROL Function
Artifact 4	Unusual Process Activity (Thread Suspension of Everything Except Thread Running in a Process Other Than Exploit Thread)
Artifact 5	Suspicious Files Written to Disk
Artifact 6	Unusual Command Line History Associated with Known CVE Techniques (CVE-2019-5736 Privilege Escalation is Visible via Unusual Command Line Commands)
Artifact 7	Suspicious File Write to System Directory Followed by Privileged Execution of File
Artifact 8	Unusual or Unexpected KERBEROS Ticket Requests
Artifact 9	Suspicious Program Running Under SYSTEM or Other Elevated Account
Artifact 10	Driver Loaded (SYSMON Event)

Artifacts Associated with Exploitation for Privilege Escalation Technique (T0890)	
Artifact 11	Network Traffic Matching Vulnerability (Snort, SURICATA)
Artifact 12	Abnormal Reads/Writes Between Processes
Artifact 13	Unusual Command Line Arguments to Application (lolbins)
Artifact 14	Artifacts Associated with Known Privilege Escalation CVEs (PE Hard Coded Debug File Path for APT28 Malware Included Reference to CVE-2014-4076 Privilege Escalation)
Artifact 15	Unusual or Unexpected Child Process Running at Elevated Privileges
Artifact 16	Execution of a Suspicious File in the System32 or Windows Directory at Privileged Level

Artifacts Associated with Remote System Discovery Technique (T0846)	
Artifact 1	Protocol Header Enumeration
Artifact 2	Protocol Content Enumeration
Artifact 3	VNC Port 5900 Calls
Artifact 4	TCP ACK Scan
Artifact 5	TCP XMAS Scan
Artifact 6	Recurring Protocol SYN Traffic
Artifact 7	TCP FIN Scans
Artifact 8	Device Failure
Artifact 9	TCP Reverse Ident Scan
Artifact 10	Sequential Protocol SYN Traffic
Artifact 11	Scans Over Industrial Network Ports with Target IPs
Artifact 12	Industrial Network Traffic Content Containing Logical Identifiers
Artifact 13	SMTP Port 25 Traffic
Artifact 14	Device Reboot
Artifact 15	Bandwidth Degradation
Artifact 16	Host Recent Connection Logs
Artifact 17	IEC 101 Traffic to Serial Devices
Artifact 18	IEC 102
Artifact 19	IEC 104
Artifact 20	OPC Network Traffic
Artifact 21	Statistical Anomalies in Network Traffic
Artifact 22	DNS Port 53 Zone Transfers
Artifact 23	Industrial Network Traffic
Artifact 24	Common Network Traffic
Artifact 25	IEC 103 Traffic (For North America)

Artifacts Associated with Remote System Discovery Technique (T0846)	
Artifact 26	IEC 61850 MMS
Artifact 27	Controller Proprietary Traffic
Artifact 28	Echo Type 8 Traffic
Artifact 29	ICMP Type 7 Traffic
Artifact 30	SNMP Port 162 Traffic
Artifact 31	SNMP Port 161 Traffic
Artifact 32	ARP Scans
Artifact 33	Operating System Queries
Artifact 34	TCP SYN Scans
Artifact 35	Industrial Network Traffic Content About Hostnames
Artifact 36	Polling Network Traffic from Unauthorized IP Sender Addresses
Artifact 37	NETBIOS Name Services Port
Artifact 38	LDAP Port
Artifact 39	Active Directory Calls
Artifact 40	Email Server Calls
Artifact 41	DNS Lookup Queries
Artifact 42	TCP Connect Scan
Artifact 43	Command Line Dialog Box Open

Artifacts Associated with Exploitation of Remote Services Technique (T0866)	
Artifact 1	SQL Protocol
Artifact 2	OPC Code Injection
Artifact 3	Vendor Specific Network Traffic
Artifact 4	Remote Network Traffic
Artifact 5	Common Network Traffic
Artifact 6	Absence of Alarm Events
Artifact 7	Alarm Events
Artifact 8	Application Logoff Event
Artifact 9	Safe Mode Reboot
Artifact 10	Blank Screens
Artifact 11	System Reboots
Artifact 12	Kernel Level Events
Artifact 13	Security Events Across Multiple Devices
Artifact 14	Host System Registry Changes

Artifacts Associated with Exploitation of Remote Services Technique (T0866)	
Artifact 15	Industrial Protocol Network Traffic
Artifact 16	Database Command Executions
Artifact 17	SMB Protocol
Artifact 18	Code Injection into the OS
Artifact 19	Application Logon Event
Artifact 20	Code Injections into Application
Artifact 21	Controller Failure
Artifact 22	Process Failure
Artifact 23	Misconfigurations of End Points
Artifact 24	Manipulation of Set Points
Artifact 25	Manipulation of Process
Artifact 26	Connection to Controller End Points
Artifact 27	Connection to Data Historian End Points
Artifact 28	Connection to EWS End Points
Artifact 29	Connection to HMI End Points
Artifact 30	Application Logs
Artifact 31	User Events Across Multiple Devices

Artifacts Associated with Loss of Availability Technique (T0826)	
Artifact 1	Process Failure Due to Loss of Required Network or System Dependency
Artifact 2	Unexplained Loss of User Data
Artifact 3	Changes In Network Routing or Usage of Redundant Control System Network Connection Due to Failed Network Path
Artifact 4	Significant Reduction or Increase in Network Traffic Due to Malware Propagation or Disappearance of Services
Artifact 5	Significant Logged Usage of Native Crypto Functions or Presence of Import of Crypto Functions in Binaries
Artifact 6	Operator or User Discovery of Encrypted or Inoperable Systems
Artifact 7	File System Modification Artifacts Might Be Associated with The Loss of Availability Might Be Present on Disk
Artifact 8	Unexplained Loss of Application Data

Artifacts Associated with Loss of View Technique (T0829)	
Artifact 1	Blocking Communications Paths and Channels (Comms, Network Infrastructure, Host Packet Routing) Might Occur via an OS or Device Modification

Artifacts Associated with Loss of View Technique (T0829)	
Artifact 2	Masked Reconfiguration of Alarm Thresholds or Other Configuration Settings Might Result in Unexpected Lack of Situational Awareness
Artifact 3	File System Modification Artifacts Might Be Associated with The Loss of View Attack Might Be Present on Disk
Artifact 4	Application Logic Hooks or Modifications Might Prevent Proper Reporting to/from an Industrial Application

APPENDIX C: OBSERVERS

This is a collection of standardized potential observers that work in operational technology organizations. It has been slightly modified by the CyOTE team from the [Job Role Groupings listed in the SANS ICS Job Role to Competency Level Poster](#) to communicate the categories of potential observers during cyber events.

Engineering  • Process Engineer • Electrical, Controls, and Mechanical Engineer • Project Engineer • Systems and Reliability Engineer • OT Developer • PLC Programmer • Emergency Operations Manager • Plant Networking • Control/Instrumentation Specialist • Protection and Controls • Field Engineer • System Integrator	Support Staff  • Remote Maintenance & Technical Support • Contractors (engineering) • IT and Physical Security Contractor • Procurement Specialist • Legal • Contracting Engineer • Insurance • Supply-chain Participant • Inventory Management/Lifecycle Management • Physical Security Specialist
Operations Technology (OT) Staff  • Operator • Site Security POC • Technical Specialists (electrical/mechanical/chemical) • ICS/SCADA Programmer	Information Technology (IT) Cybersecurity  • ICS Security Analyst • Security Engineering and Architect • Security Operations • Security Response and Forensics • Security Management (CSO) • Audit Specialist
Operational Technology (OT) Cybersecurity  • OT Security • ICS/SCADA Security	• Security Tester
Management  • Plant Manager • Risk/Safety Manager • Business Unit Management • C-level Management	Information Technology (IT) Staff  • Networking and Infrastructure • Host Administrator • Database Administrator • Application Development • ERP/MES Administrator • IT Management

REFERENCES

- ¹ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ² [Wired | Paul Boutin | “Slammed!” | <https://www.wired.com/2003/07/slammer/> | 1 July 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³ [Microsoft | “Win32/Slammer” | <https://www.microsoft.com/en-us/wdsi/threats/malware-encyclopedia-description?name=Win32%2FSlammer> | 20 March 2009 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ⁴ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ⁵ [Carnegie Mellon University CERT Division | “2003 CERT Advisories” | https://resources.sei.cmu.edu/asset_files/WhitePaper/2003_019_001_496200.pdf | 25 January 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁶ [Cisco Systems, Inc. | “Combating Internet Worms “SQL Slammer”” | <https://www.cisco.com/web/offer/powernow/docs/security/Slammer.pdf> | 1 February 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁷ [Cisco Systems, Inc. | “Combating Internet Worms “SQL Slammer”” | <https://www.cisco.com/web/offer/powernow/docs/security/Slammer.pdf> | 1 February 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁸ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁹ [ZDNET | Robert Lemos | “Counting the cost of Slammer” | <https://www.zdnet.com/article/counting-the-cost-of-slammer/> | 3 February 2003 | Accessed on 18 January 2023 | The source is publicly available information and does not contain classification markings]
- ¹⁰ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹¹ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹² [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹³ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

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- ¹⁴ [Microsoft | “Microsoft Security Bulletin MS02-039 - Critical” | <https://learn.microsoft.com/en-us/security-updates/SecurityBulletins/2002/ms02-039?redirectedfrom=MSDN> | 24 July 2002 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁵ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁶ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁷ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁸ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ¹⁹ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²⁰ [United States Nuclear Regulatory Commission | Edward Markey | “EDO Principal Correspondence Control” | <https://www.nrc.gov/docs/ML0329/ML032970134.pdf> | 22 October 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²¹ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²² [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²³ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²⁴ [Carnegie Mellon University CERT Division | “2003 CERT Advisories” | https://resources.sei.cmu.edu/asset_files/WhitePaper/2003_019_001_496200.pdf | 25 January 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²⁵ [The Register | John Leyden | “SQL worm slams the Net” | https://www.theregister.com/2003/01/27/sql_worm_slams_the_net/ | 27 January 2003 | Accessed on 18 January 2023 | The source is publicly available information and does not contain classification markings]
- ²⁶ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

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- ²⁷ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²⁸ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ²⁹ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³⁰ [Westinghouse Nuclear | “Plant Process Computer System” | <https://www.westinghousenuclear.com/Portals/0/operating%20plant%20services/automation/plant%20computer%20systems/NA-0029%20Plant%20Process%20Computers.pdf> | 1 July 2017 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³¹ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³² [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³³ [United States Nuclear Regulatory Commission | Edward Markey | “EDO Principal Correspondence Control” | <https://www.nrc.gov/docs/ML0329/ML032970134.pdf> | 22 October 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³⁴ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³⁵ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³⁶ [Microsoft | “Microsoft Security Bulletin MS02-039 - Critical” | <https://learn.microsoft.com/en-us/security-updates/SecurityBulletins/2002/ms02-039?redirectedfrom=MSDN> | 24 July 2002 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³⁷ [Microsoft | “Microsoft Statement on the “Slammer” Worm Attack | <https://news.microsoft.com/2003/01/25/microsoft-statement-on-the-slammer-worm-attack/> | 25 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ³⁸ [Carnegie Mellon University CERT Division | “2003 CERT Advisories” | https://resources.sei.cmu.edu/asset_files/WhitePaper/2003_019_001_496200.pdf | 25 January 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ³⁹ [Carnegie Mellon University CERT Division | “2002 CERT Advisories” | https://resources.sei.cmu.edu/asset_files/WhitePaper/2002_019_001_496196.pdf | 29 July 2002 |

Accessed on 21 December 2002 | The source is publicly available and does not contain classification markings]

⁴⁰ [Carnegie Mellon University CERT Coordination Center | Allen Householder | “CERT Advisory CA-2003-04 MS-SQL Server Worm” | <https://vuls.cert.org/confluence/display/historical/CERT+Advisory+CA-2003-04+MS-SQL+Server+Worm> | 25 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁴¹ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁴² [Microsoft | “Microsoft Security Bulletin MS02-039 - Critical” | <https://learn.microsoft.com/en-us/security-updates/SecurityBulletins/2002/ms02-039?redirectedfrom=MSDN> | 24 July 2002 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁴³ [Threat Post | David Litchfield | “The Inside Story of SQL Slammer” | <https://threatpost.com/inside-story-sql-slammer-102010/74589/> | 20 October 2010 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁴⁴ [National Institute of Standards and Technology | “National Vulnerability Database: CVE-2002-0649 Detail” | <https://nvd.nist.gov/vuln/detail/CVE-2002-0649> | 12 August 2002 | Accessed on 21 December 2022 | The source is publicly available information and does not contain any classification markings]

⁴⁵ [Threat Post | David Litchfield | “The Inside Story of SQL Slammer” | <https://threatpost.com/inside-story-sql-slammer-102010/74589/> | 20 October 2010 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁴⁶ [Wired | Paul Boutin | “Slammed!” | <https://www.wired.com/2003/07/slammer/> | 1 July 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁴⁷ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁴⁸ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁴⁹ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁵⁰ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁵¹ [United States Nuclear Regulatory Commission | Edward Markey | “EDO Principal Correspondence Control” | <https://www.nrc.gov/docs/ML0329/ML032970134.pdf> | 22 October 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁵² [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

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- ⁵³ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁵⁴ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ⁵⁵ [Wired | Paul Boutin | “Slammed!” | <https://www.wired.com/2003/07/slammer/> | 1 July 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁵⁶ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁵⁷ [Cisco Systems, Inc. | “Combating Internet Worms “SQL Slammer”” | <https://www.cisco.com/web/offer/powernow/docs/security/Slammer.pdf> | 1 February 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁵⁸ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ⁵⁹ [Symantec | Douglas Knowles | “W32.SQLExp.Worm” | <https://web.archive.org/web/20061205015822/http://sarc.com/avcenter/venc/data/w32.sqlexp.worm.html> | 24 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ⁶⁰ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ⁶¹ [Wired | Paul Boutin | “Slammed!” | <https://www.wired.com/2003/07/slammer/> | 1 July 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁶² [Wired | Paul Boutin | “Slammed!” | <https://www.wired.com/2003/07/slammer/> | 1 July 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁶³ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]
- ⁶⁴ [Microsoft | “Microsoft Security Bulletin MS02-039 - Critical” | <https://learn.microsoft.com/en-us/security-updates/SecurityBulletins/2002/ms02-039?redirectedfrom=MSDN> | 24 July 2002 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁶⁵ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]
- ⁶⁶ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

[SQLExp.pdf](#) | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁶⁷ [MITRE | “Native API” | <https://attack.mitre.org/techniques/T0834/> | 13 April 2021 | Accessed on 8 March 2023 | The source is publicly available information and does not contain classification markings]

⁶⁸ [Threat Post | David Litchfield | “The Inside Story of SQL Slammer” | <https://threatpost.com/inside-story-sql-slammer-102010/74589/> | 20 October 2010 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁶⁹ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁷⁰ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁷¹ [Wired | Paul Boutin | “Slammed!” | <https://www.wired.com/2003/07/slammer/> | 1 July 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁷² [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁷³ [Wired | Paul Boutin | “Slammed!” | <https://www.wired.com/2003/07/slammer/> | 1 July 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁷⁴ [Threat Post | David Litchfield | “The Inside Story of SQL Slammer” | <https://threatpost.com/inside-story-sql-slammer-102010/74589/> | 20 October 2010 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁷⁵ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁷⁶ [Microsoft | “Microsoft Security Bulletin MS02-039 - Critical” | <https://learn.microsoft.com/en-us/security-updates/SecurityBulletins/2002/ms02-039?redirectedfrom=MSDN> | 24 July 2002 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁷⁷ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁷⁸ [University of California and USC Information Science Institute | Mohit Lad, Xiaoliang Zhao, and others | “Analysis of BGP Update Surge during Slammer Worm Attack” | 2003 | Accessed on 21 December 2002 | The source is publicly available and does not contain classification markings]

⁷⁹ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁸⁰ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

[SQLExp.pdf](#) | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁸¹ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁸² [Caida | David Moore, Vern Paxson, and others | “The Spread of the Sapphire/Slammer SQL Worm” | <https://archive.nanog.org/meetings/nanog27/presentations/worm.pdf> | 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁸³ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁸⁴ [F-Secure | “Worm:W32/Slammer” | <https://www.f-secure.com/v-descs/mssqlm.shtml> | 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁸⁵ [Caida | David Moore, Vern Paxson, and others | “The Spread of the Sapphire/Slammer SQL Worm” | <https://archive.nanog.org/meetings/nanog27/presentations/worm.pdf> | 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁸⁶ [Carnegie Mellon University CERT Division | “2003 CERT Advisories” | https://resources.sei.cmu.edu/asset_files/WhitePaper/2003_019_001_496200.pdf | 25 January 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁸⁷ [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁸⁸ [University of California and USC Information Science Institute | Mohit Lad, Xiaoliang Zhao, and others | “Analysis of BGP Update Surge during Slammer Worm Attack” | 2003 | Accessed on 21 December 2002 | The source is publicly available and does not contain classification markings]

⁸⁹ [F-Secure | “Worm:W32/Slammer” | <https://www.f-secure.com/v-descs/mssqlm.shtml> | 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁹⁰ [Symantec | Jensenne Roculan, Sean Hittel, and others | “SQLExp SQL Server Worm Analysis” | <https://web.archive.org/web/20170424202953/http://securityresponse.symantec.com/avcenter/Analysis-SQLExp.pdf> | 28 January 2003 | Accessed on 21 December 2022 | The source is publicly available and does not contain classification markings]

⁹¹ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁹² [IEEE Computer Society | David Moore, Vern Paxson, and others | “Inside the Slammer Worm” | <https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=1219056> | 11 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁹³ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁹⁴ [Cisco Systems, Inc. | “Combating Internet Worms “SQL Slammer”” | <https://www.cisco.com/web/offer/powernow/docs/security/Slammer.pdf> | 1 February 2003 | Accessed on

21 December 2022 | The source is publicly available information and does not contain classification markings]

⁹⁵ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁹⁶ [The Register | Kevin Poulson | “Slammer worm crashed Ohio nuke plant net” | https://www.theregister.com/2003/08/20/slammer_worm_crashed_ohio_nuke/ | 20 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁹⁷ [United States Nuclear Regulatory Commission | “NRC INFORMATION NOTICE 2003-14: POTENTIAL VULNERABILITY OF PLANT COMPUTER NETWORK TO WORM INFECTION ” | <https://www.nrc.gov/docs/ML0324/ML032410430.pdf> | 29 August 2003 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]

⁹⁸ [Westinghouse Nuclear | “Plant Process Computer System” | <https://www.westinghousenuclear.com/Portals/0/operating%20plant%20services/automation/plant%20computer%20systems/NA-0029%20Plant%20Process%20Computers.pdf> | 1 July 2017 | Accessed on 21 December 2022 | The source is publicly available information and does not contain classification markings]